

Anant Bhardwaj

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[GitHub](#) | [LeetCode](#) | [Kaggle](#) | [Portfolio](#)

EDUCATION

Indian Institute of Information Technology Manipur

Imphal, India

Bachelor of Technology in Computer Science & Technology; **CGPA: 8.68/10**

Aug. 2024 – May 2028

- Active participant in competitive programming and technical project development initiatives

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS

ML & Data Science: Machine Learning (Supervised & Unsupervised), Computer Vision, NLP, Exploratory Data Analysis (EDA), Feature Engineering, Model Deployment

Libraries & Frameworks: scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Seaborn, FastAPI, Streamlit, Raylib

Tools & Databases: Git/GitHub, Linux, PostgreSQL, MySQL, Redis, Jupyter, VS Code

Core CS Concepts: Data Structures & Algorithms (DSA), Machine Learning, MLOps, Object-Oriented Programming (OOP), DBMS, Operating Systems, Data Pipelines

PROJECTS

AI Learning Path Generator | *Next.js, Python, FastAPI, PostgreSQL, Redis, Gemini, React Flow*

- Engineered a full-stack application utilizing the Gemini API to dynamically generate structured learning roadmaps (Directed Acyclic Graphs), featuring an interactive React Flow UI with actionable milestones and progress tracking.
- Optimized AI API costs and system performance by implementing a robust data pipeline, leveraging Redis for efficient caching and PostgreSQL for permanent roadmap storage.

Wrong Direction Vehicle Detection | *Python, Computer Vision, Data Pipelines*

- Engineered a Python-based computer vision pipeline to detect and track vehicles traveling in the wrong direction.
- Developed robust backend modules including detection, tracking, and direction evaluation scripts to process dynamic video feeds.
- Built an interactive dashboard application to visualize the detection pipeline and output final processed tracking videos.

Rocket Simulator | *C++, Raylib, Physics Simulation, OOP*

- Developed a physics-based rocket simulator in C++, implementing core aerodynamic and astrodynamics equations to accurately model trajectory and propulsion mechanics.
- Designed a robust object-oriented architecture to manage system states and simulation data, optimizing computational performance for real-time physics calculations.

Ahouba.com (Official Techfest Website) | *React.js, Three.js, GSAP, Blender*

- Collaborated to develop IIIT Manipur's official techfest website, integrating Three.js and Blender-optimized custom 3D models to build an immersive, "game-like" environment with basic physics and character controls.
- Enhanced the 2D interface using React.js and GSAP for high-performance animations to ensure seamless transitions with the 3D canvas, while managing version control and deployment via Git/GitHub.

2D Space Game (Web-Based) | *JavaScript, HTML5 Canvas, CSS*

- Developed an interactive 2D space dodge game with real-time collision detection and a dynamic scoring system, designing object-oriented mechanics via JavaScript classes for player, obstacle, and state management.
- Implemented an efficient game loop architecture and optimized the rendering pipeline to handle multiple moving objects, ensuring sustained 60 FPS performance, smooth gameplay, and responsive keyboard controls.

Event-Driven Route Optimization System | *GDG on Campus Solution Challenge*

- Designed and engineered an event-driven route optimization system as a comprehensive project submission for the 2025 GDG on Campus Solution Challenge.
- Applied core software engineering fundamentals and algorithmic concepts to effectively model, optimize, and solve complex real-world logistical routing problems.

EXTRA CURRICULAR & HACKATHONS

Pathway x IIT Ropar | *Hackathon Participant*

- Engaged in intense algorithmic and development challenges, practicing code optimization and technical documentation in a highly competitive team setting.

Ethos 2025, IIT Guwahati | *Hackathon Participant*

- Collaborated with a team to build scalable software solutions under tight time constraints, focusing on agile development principles and rapid prototyping.

ADDITIONAL DETAILS

- Solved **451+ algorithmic problems on LeetCode (Max Rating: 1510)**, focusing on Hash Tables, Dynamic Programming, Binary Search, and Greedy Algorithms to optimize time and space complexities.
- Solved **102+ algorithmic problems on Codeforces**, focusing on data structures, number theory, and implementation-heavy problems to strengthen competitive programming fundamentals.
- Consistently maintain strong academic performance (**8.68 CGPA**) while building and deploying multiple full-stack web projects.
- Active contributor in the college technical community, mentoring peers in programming concepts and collaborating on open-source codebases.